Sunol FlowLab VR

Hybrid Jar Test Bench and Hero Observation Tank Design Brief

Proposed repository file: `docs/DESIGN_DIRECTION_JAR_TEST_HYBRID.md`
Status: Approved design direction, staged implementation
Applies to: Product presentation, visual hierarchy, interaction meaning, instrumentation, rendering, and future roadmap
Execution authority: `IMPLEMENTATION_PLAN.md` and the individual batch plans

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1. Purpose

Sunol FlowLab VR should retain the instantly recognizable visual language of a traditional six-jar test without limiting the experience to a literal recreation of six small jars on a laboratory bench.

The classic jar test provides authenticity and immediate recognition for water-treatment professionals. However, six small jars alone do not provide enough visual scale, drama, or process readability for the central VR experience.

The approved direction is:

> Use a classic six-jar test bench as the recognizable control and comparison apparatus, paired with one larger hero observation tank that presents the selected experiment with greater visual clarity and impact.

The jars establish what the experiment is.

The hero tank shows why the experiment matters.

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2. Product Direction

The user stands at a stylized water-treatment experiment station containing:

- a recognizable six-jar test rack;
- one larger central observation tank;
- a physical relative-dose control;
- a deliberate Start control;
- restrained physical instrumentation;
- a mounted dose-response plot;
- a fictionalized water-treatment environment.

The experience is inspired by a jar test but is not intended to be a literal commercial jar-test simulator.

It is a stylized educational instrument designed to make qualitative treatment behavior easier to understand.

The experience remains:

- phenomenological;
- deterministic;
- fictionalized;
- educational;
- non-calibrated;
- unsuitable for operational guidance or chemical-dose prediction.

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3. Plan Authority

This document defines:

- product intent;
- visual hierarchy;
- interaction meaning;
- apparatus composition;
- source-of-truth requirements;
- sensory and presentation constraints.

It does not independently schedule implementation.

`IMPLEMENTATION_PLAN.md` and the individual batch plans remain authoritative for:

- implementation order;
- blockers;
- scope;
- tests;
- evidence;
- acceptance criteria;
- batch completion.

When this document and a batch plan appear to disagree about implementation timing or scope, the batch plan governs.

The authority hierarchy is:

```
CLAUDE.md and architecture contracts
        ↓
IMPLEMENTATION_PLAN.md
        ↓
individual batch plan
```

```
        ↓
this design-direction brief
        ↓
implementation
```

The affected batch plans should reference this document directly.

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4. Core Experience

The hero observation tank is the primary live visualization.

It should make the following behavior easy to see:

- initial raw-water haze;
- rapid mixing;
- flocculation;
- visible floc growth;
- larger floc settling faster;
- development of a clearing front;
- residual suspended material;
- the difference between underdose, near-optimum dose, and overdose.

The six jars support this experience rather than competing with it.

The user should understand the relationship as:

```
classic jar-test apparatus
        ↓
select a relative dose condition
        ↓
observe the treatment process at a larger readable scale
        ↓
measure and compare the result
```

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5. Role of the Six-Jar Apparatus

The six-jar apparatus has three purposes:

1. authenticity;
2. canonical dose selection;
3. simplified comparison.

It is not the complete experiment history.

5.1 Authenticity anchor

The jars should immediately suggest a jar test to water-treatment professionals.

They should resemble the organization and purpose of a traditional jar-test bench without reproducing a specific commercial product, agency setup, or sensitive real facility.

The apparatus should communicate:

- relative chemical-dose comparison;
- controlled experimentation;
- treatment observation;
- comparison of final outcomes.

5.2 Canonical dose presets

The six jars represent canonical relative-dose presets:

Jar	Relative dose
1	0
2	2
3	4
4	6
5	8
6	10

The application must continue to support the complete integer `DoseIndex` range from `0` through `10`.

The six jars do not replace the 11-detent dose contract.

Intermediate values such as `1`, `3`, `5`, `7`, and `9` remain selectable through the primary physical dose control.

The interface must communicate that the jars are convenient canonical comparison presets, not storage locations for every possible dose.

5.3 Canonical comparison summaries

A completed trial may update a simplified static summary in a jar only when its dose exactly matches that jar's canonical preset.

Examples include:

- final relative haze;
- simplified settled-material appearance;
- a tested/not-tested indicator;
- a small result token;
- a restrained clarity gradient.

Canonical jar summaries are optional physical comparisons. They are not separate simulations.

They must not contain:

- their own simulation clock;
- continuously moving particles;
- independent turbidity calculations;
- per-frame process logic;
- renderer-derived treatment outcomes.

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6. Complete Experiment Memory

The mounted dose-response plot and the versioned experiment log are the sole complete records of experiment history.

They must support all integer dose values from `0` through `10`.

A completed dose-3 trial should:

- remain visible in the hero tank at completion;
- update the measurement gauge;
- add a point to the complete plot;
- persist in the experiment log;
- leave the six canonical jar summaries unchanged.

A completed dose-4 trial should:

- remain visible in the hero tank at completion;
- update the measurement gauge;
- add a point to the complete plot;
- persist in the experiment log;
- update the dose-4 canonical jar summary once.

The jars must not be described as the complete experiment memory.

Preferred terminology:

- **canonical jars**

- **canonical comparison samples**
- **dose presets**
- **comparison summaries**

Avoid describing them as eleven-dose storage.

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7. Role of the Hero Observation Tank

The hero observation tank is the authoritative live process presentation.

It should:

- display the currently selected dose trial;
- provide enough visual scale to see floc formation;
- make vertical settling and the clearing front readable;
- support the measurement event;
- remain visually dominant over the jars, gauges, plot, and environment.

The hero tank does not represent a literal full-scale process basin.

It is a stylized observation chamber inspired by jar-test behavior.

Its increased scale exists for readability and visual storytelling, not to imply plant-scale operation, calibrated treatment performance, or dose prediction.

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8. Visual Hierarchy

The scene should follow this hierarchy:

1. hero observation tank;
2. active dose control and selected canonical jar;
3. floc and clearing-front behavior;
4. measurement gauge and result plot;
5. remaining canonical jar summaries;
6. background environment.

The jar rack should communicate context and comparison without overpowering the live process.

The environment must remain subordinate to the apparatus.

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9. Source-of-Truth Rule

The canonical jars, hero tank, gauge, plot, nephelometer, measurement cue, audio, and haptic feedback must derive from the same authoritative application and simulation state.

Approved data flow:

```
validated dose command
        ↓
authoritative deterministic simulation
        ↓
authoritative turbidity-band and trial-result records
        ↓
hero tank presentation
canonical jar summary
gauge
plot
measurement cue
audio response
```

The following are prohibited:

- renderer-only dose outcomes;
- jar clarity calculated separately from the completed result;
- hero-tank behavior that disagrees with the result record;
- visual effects that imply successful flocculation solely because a phase began;
- independent turbidity calculations inside instruments;
- decorative “best jar” effects unsupported by trial data;
- audio or lighting that communicates a better result than the simulation produced.

Visual transforms may improve readability, but they must consume authoritative outputs.

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10. Simulation and Floc Presentation

Actual deterministic floc aggregation remains the preferred behavior.

When particles or floc structures merge:

1. the simulation updates authoritative mass, size, position, velocity, density, and active-slot state;
2. the renderer may apply a short visual tween to make the merge readable;
3. settling behavior follows the authoritative updated floc properties.

The renderer must not enlarge, recolor, or settle particles merely because the flocculation or settling phase is active.

Visual animation may clarify an event that occurred.

It may not invent the event.

The intended educational chain is:

```
dose affects collision success
        ↓
successful collisions create larger floc
        ↓
larger floc settles faster
        ↓
suspended material decreases
        ↓
clarity improves
```

Low dose and excessive dose must produce visibly poorer results than the near-optimum condition.

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11. Runtime Ownership Correction

Before additional treatment behavior is implemented, complete the existing runtime-ownership correction.

Required ownership

- `src/app` owns simulation lifecycle and orchestration.
- `src/sim` owns deterministic process state and behavior.
- `src/render` consumes externally owned read-only simulation views.
- `src/xr` emits discrete validated commands.
- React does not own continuously changing process state.

Required outcomes

- rendering does not instantiate the simulation clock;
- rendering does not start or advance the simulation;
- rendering does not reset authoritative process state;
- rendering does not import application-owned lifecycle or telemetry modules;
- the runtime can start, pause, reset, and advance without React or WebGL;
- deterministic headless stepping remains supported;
- no generalized engine, event bus, or global state framework is introduced.

Required tests

1. **Import-boundary tests**

2. forbid `src/render` imports from `src/app`;

3. forbid browser, React, Three.js, and XR imports from `src/sim`.

4. **Runtime lifecycle tests**

5. runtime creates and owns process state;

6. runtime starts, pauses, resets, and advances deterministically.

7. **Renderer contract tests**

8. renderer consumes externally provided state;

9. renderer does not instantiate `FixedStepClock`;
10. renderer does not call authoritative simulation-step functions.

11. **Headless behavior tests**

12. simulation state advances without React, WebGL, or a browser.

13. **Optional development diagnostics**

14. record unexpected React rerenders during a running trial.

An absolute zero-render count is not the primary gate. The enforced invariant is that continuously changing simulation state does not require React ownership or React state updates.

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12. Static Canonical Jar Summaries

Canonical jar summaries should use a write-on-completion model.

When a completed result matches a canonical dose:

1. the application identifies the corresponding jar;
2. it derives a documented display value from the authoritative result;
3. it updates the jar material or static display state once;
4. it stores the result ID associated with the summary;
5. no continuous jar simulation is created.

A suitable structure may resemble:

```
interface CanonicalJarSummary {
  readonly dose: 0 | 2 | 4 | 6 | 8 | 10
  readonly trialId: string
```

```
  readonly endpointTurbidity: number
  readonly displayClarity: number
}
```

The display transform must consume the authoritative completed result.

Canonical jar summaries must be:

- cleared consistently with experiment-log clearing;
- restored consistently with persistence;
- traceable to a completed result;
- updated only on trial completion;
- static between result updates.

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13. Nephelometer and Measurement Presentation

The measurement event should use a recognizable emitter-and-detector arrangement around the sample zone.

The initial implementation should prefer:

- a narrow emissive beam or illuminated sample line;
- a brief detector glow;
- a physical gauge response;
- a restrained measurement sound;
- intensity mapped from the authoritative turbidity sample.

Avoid an expensive full volumetric beam or large transparent cone in the first implementation.

A more atmospheric scattering effect may be evaluated later only after:

- the hero tank is visually successful;
- physical Quest performance is measured;
- the transparency budget remains healthy;
- the tank, gauge, plot, and floc remain readable.

The nephelometer should reinforce the lesson, not become the dominant spectacle.

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14. Rendering and Performance Rules

14.1 Transparency budget

The primary rendering risk is expected to be layered transparency and screen-space overdraw rather than particle count alone.

Therefore:

- prefer opaque or near-opaque instanced particles;
- use one turbidity gradient surface rather than many transparent layers;
- avoid realistic refraction;
- avoid stacked transparent jar walls where a cheaper presentation works;
- avoid multiple full-tank blended effects;
- require before-and-after Quest measurements for large transparent effects.

No new large transparent effect should be accepted without evidence showing:

- average frame time;
- p95 frame time;
- draw-call change;
- memory impact where measurable;
- no material loss of label, gauge, plot, floc, or clearing-front readability.

14.2 React ownership

React may handle:

- application composition;
- discrete phase transitions;
- XR session lifecycle;
- completed-trial updates;
- low-frequency accessibility state;
- development-only configuration surfaces.

React must not own:

- per-particle state;
- continuously changing turbidity arrays;
- simulation-clock progression;
- floc motion;
- clearing-front animation;
- continuously updated gauge position.

Continuously changing visuals should read from stable runtime objects, typed arrays, snapshots, or refs.

14.3 Simulation thread

The deterministic simulation remains on the main thread unless measured evidence shows it has become a meaningful bottleneck.

A Web Worker is not part of the current plan.

14.4 XR quality controls

Foveation and framebuffer scale are configuration and measurement tools, not unconditional optimizations.

The pinned Three.js version already defaults XR foveation to its maximum setting.

Any explicit setting should be recorded for reproducibility.

Framebuffer scale may be adjusted before session entry only when physical-device evidence shows it is necessary.

Any reduction must be evaluated against:

- jar labels;
- gauge readability;
- plot readability;
- floc visibility;
- clearing-front visibility;
- controller interaction accuracy.

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15. Interaction Model

15.1 Primary dose control

A physical 11-detent lever is preferred.

A documented rotary-wheel fallback may replace it if headset testing shows the lever is unreliable.

The control must emit only integer `DoseIndex` values from `0` through `10`.

15.2 Canonical jar selection

Selecting a canonical jar may:

- move or suggest the corresponding dose preset;
- highlight the matching dose on the physical control;
- load a completed canonical result for review;
- highlight the corresponding plotted result;
- prepare the hero tank for that dose.

Jar selection must not bypass the validated dose-command contract.

15.3 Start control

A separate deliberate physical Start control begins the live hero-tank trial.

It should emit exactly one Start command per deliberate activation.

15.4 Experiment actions

Refill/reset and experiment-log clearing remain separate actions.

Refill must not clear experiment history.

Clearing the experiment log must also clear visible plotted results and canonical jar summaries according to the persistence policy.

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16. Haptics and Audio

16.1 Haptics

Haptics should be implemented with the physical controls because they influence interaction quality.

Approved uses include:

- a light pulse when crossing into a dose detent;
- a stronger pulse for a confirmed Start press;
- optional feedback for refill or clear actions.

Haptics must be:

- capability checked;
- sparse;
- event driven;
- optional;
- safely ignored when unsupported.

Haptic state must not enter the simulation domain.

16.2 Audio

Audio remains primarily part of the planned presence and audio batch.

Earlier batches may define no-op event hooks for:

- dose-detent tick;
- Start press;
- mixer start and stop;

- phase transitions;
- measurement capture;
- refill;
- experiment-log clearing.

Potential final audio includes:

- restrained mixer hum;
- subtle water movement;
- mechanical detent tick;
- measurement beep;
- quiet settling ambience.

The experience must remain understandable when muted.

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17. Environment and Lighting

The apparatus remains the visual hero.

The initial environment should use:

- restrained low-poly geometry;
- simple shared materials;
- limited real-time lights;
- no dynamic shadows;
- near-field objects that establish scale and parallax;
- only the plant context visible from the experiment station.

A panorama or environment map may be evaluated later.

It should not be added solely to make the scene appear more realistic.

Before adoption, it must pass:

- load-time review;
- decoded-memory review;
- Quest performance review;
- label-contrast review;
- apparatus-dominance review;
- public-data and asset-provenance review.

The visual identity is a stylized educational installation inspired by drinking-water treatment, not a recreation of a specific facility.

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18. Recognition and Comprehension Validation

The recognition criterion requires outside evidence.

Once the bench and hero-tank composition is blocked out, show one unlabeled screenshot to:

- at least one water-treatment operator or educator;
- preferably one person without water-treatment experience.

Ask:

1. “What do you think this apparatus represents?”
2. “What would you expect to do here?”

Do not explain the project before they answer.

Record the responses in:

```
docs/UX_VALIDATION.md
```

The composition should be reconsidered before instrumentation is finalized when:

- the operator does not recognize the jar-test connection;
- the non-operator cannot identify it as a comparative experiment;
- the jars visually overpower the hero tank;
- the relationship between the jars and hero tank is unclear;
- the user expects six simultaneous live simulations when only one is authoritative.

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19. Implementation Mapping

The hybrid direction maps to the existing batch plan as follows:

Design work	Governing batch
Runtime ownership correction	Batch 01A closure
Hero tank and jar-rack composition blockout	Batch 03
Dose control, Start control, haptics, and optional jar presets	Batch 04
Simulation and XR integration	Batch 05
Complete treatment-cycle behavior	Batch 06
Canonical jar summaries, plot, gauge, nephelometer, and persistence	Batch 07

Design work	Governing batch
Headset readability and clearing-front refinement	Batch 08
Desktop spectator presentation	Batch 09
Audio, lighting, environment, and measurement-effect polish	Batch 10
Approved export and sharing features	Batch 11

Each affected batch plan should include:

> This batch must also follow `docs/DESIGN_DIRECTION_JAR_TEST_HYBRID.md`. This design brief governs product intent and presentation meaning. The batch plan remains authoritative for implementation timing, scope, tests, evidence, and acceptance.

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20. Staged Implementation

Stage 1: Close the foundation

Complete:

- runtime ownership correction;
- dependency-boundary tests;
- deterministic simulation foundation;
- performance and memory evidence;
- status and plan synchronization.

Stage 2: Prove the phenomenon

Complete:

- dose-response behavior;
- deterministic floc formation;
- settling;
- turbidity-band authority;
- one compelling hero-tank desktop proof;
- 11-dose headless sweep;
- low, optimum, and high comparison;
- blocked-out jar-rack composition;
- recognition validation.

The canonical jars may remain static geometry during this stage.

Stage 3: Prove XR interaction

Implement:

- 11-detent lever or rotary fallback;
- Start control;
- detent haptics;
- seated and standing ergonomics;
- left- and right-hand use;
- optional canonical jar preset selection;
- stable integer command emission.

Stage 4: Integrate the treatment cycle

Connect:

- selected dose;
- authoritative runtime;
- hero tank;
- treatment phases;
- measurement;
- result capture;
- reset/refill;
- XR interaction state.

Stage 5: Add instrumentation and memory

Implement:

- static canonical jar summaries;
- gauge;
- complete dose-response plot;
- nephelometer geometry;
- experiment-log persistence;
- refill control;
- physical clear-history action.

Stage 6: Add presence and polish

Evaluate:

- final haptic tuning;
- spatial audio;
- restrained measurement-light effect;
- environment lighting;
- optional panorama or environment map;
- material and composition polish.

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21. Version 1 Boundary

Version 1 includes:

- one live authoritative simulation;
- one hero observation tank;
- one six-jar canonical preset rack;
- one 11-detent relative-dose control;
- one treatment cycle at a time;
- static canonical comparison summaries;
- a complete dose-response plot;
- persistent experiment history;
- desktop spectator mode;
- Quest VR mode.

Version 1 does not require:

- six concurrent full simulations;
- a universal multi-vessel engine;
- calibrated chemical-dose units;
- pH chemistry;
- temperature control;
- hand tracking;
- realistic volumetric lighting;
- fluid simulation;
- CFD;
- operational recommendations;
- a full plant walkthrough.

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22. Future Product Direction

The six-jar rack provides a natural future path without forcing the current architecture to support it prematurely.

Potential v2 headline:

> A comparative six-jar experiment where multiple dose conditions can be reviewed together using the proven single-vessel treatment model.

Possible later research includes:

- six simultaneously active jar simulations;
- relative pH as a second educational variable;
- different fictionalized raw-water presets;
- guided classroom mode;

- JSON or PNG result export;
- shareable seed URLs;
- hand-tracking interaction;
- multi-user demonstration modes.

These remain roadmap or research concepts, not current implementation commitments.

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23. Acceptance Criteria

The hybrid direction is successful when:

- a water-treatment operator recognizes the jar-test connection from an unlabeled image;
- a non-operator identifies the setup as a comparative experiment;
- the hero tank remains the most compelling and readable element;
- the jars read as canonical presets rather than complete history;
- underdose, near-optimum dose, and overdose are visibly distinguishable;
- floc growth and settling derive from authoritative simulation behavior;
- canonical jar summaries agree with completed trial results;
- odd-dose results remain visible in the complete plot and experiment log;
- gauge, water appearance, measurement cue, plot, and persisted result agree through the authoritative turbidity record;
- the apparatus remains usable seated and standing;
- physical interaction produces only validated integer dose commands;
- no continuous process state requires React ownership;
- the experience sustains the required physical Quest performance target;
- no visual effect implies calibrated plant prediction;
- the six-jar apparatus does not require six live simulations or a universal engine.

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24. Final Design Decision

Sunol FlowLab VR will not be a literal six-jar simulator and will not abandon the iconic jar-test image.

It will use a hybrid interpretive apparatus:

> A classic six-jar test bench provides authenticity and canonical dose selection. One larger hero observation tank magnifies the selected treatment process into a visually compelling educational experience. The mounted plot and persisted experiment log provide the complete history across all eleven dose values, while the jars provide static summaries only for their six canonical presets.

This direction strengthens the project's visual identity while preserving:

- one authoritative simulation;
- clear architecture boundaries;

- deterministic behavior;
- a controlled rendering budget;
- a realistic v1 scope;
- a natural path for future expansion.